The types of temporary differences that give rise to significant portions of the Company's deferred tax assets and liabilities are as follows (in thousands):

	Year ended December 31,	
	2019	2018
Deferred tax assets		
Net operating loss and credit carry-forwards	$ 196,930	$ 109,812
Research and development credits	24,452	16,380
Sales tax liability	157	258
Share-based compensation	5,937	5,435
Accrued liabilities	6,612	5,135
Gross deferred tax assets	234,088	137,020
Valuation allowance	(180,090)	(94,118)
Total deferred tax assets	53,998	42,902
Deferred tax liabilities		
Convertible debt discount	(16,701)	(21,035)
Deferred sales commissions	(28,601)	(18,253)
Acquired intangibles	(3,857)	(2,670)
Property and equipment	(6,731)	(3,573)
Net deferred tax (liabilities) assets	$ (1,892)	$ (2,629)

On December 22, 2017, the Tax Cuts and Jobs Act of 2017 (the "Act") was signed into law making significant changes to the Internal Revenue Code. Changes include, but are not limited to, a corporate tax rate decrease from 35% to 21% effective for tax years beginning after December 31, 2017, the transition of U.S. international taxation from a worldwide tax system to a territorial system, and a one-time transition tax on the mandatory deemed repatriation of cumulative foreign earnings as of December 31, 2017.

In connection with the acquisition of Connect First on January 14, 2019, a net deferred tax liability of $3.2 million was established, the most significant component of which is related to the book/tax basis differences associated with the acquired technology and customer relationships. The net deferred tax liability from this acquisition created an additional source of income to realize deferred tax assets. As the Company continues to maintain a full valuation allowance against its deferred tax assets, this additional source of income resulted in the release of the Company's previously recorded valuation allowance against deferred assets. Consistent with the applicable guidance the release of the valuation allowance of $3.2 million caused by the acquisition was recorded in the consolidated financial statements outside of acquisition accounting as a tax benefit to the Consolidated Statements of Operations.

As of December 31, 2019, the Company has federal net operating loss carryforwards of approximately $782.7 million, of which approximately $272.9 million expire between 2023 and 2037 and the remainder do not expire. As of December 31, 2019, the Company had state net operating loss carryforwards of approximately $675.6 million which will begin to expire in 2021. The Company also has research credit carryforwards for federal and California tax purposes of approximately $20.2 million and $15.7 million, respectively, available to reduce future income subject to income taxes. The federal research credit carryforwards will begin to expire in 2028 and the California research credits carry forward indefinitely.

The Internal Revenue Code of 1986, as amended, imposes restrictions on the utilization of net operating losses in the event of an "ownership change" of a corporation. Accordingly, a company's ability to use net operating losses may be limited as prescribed under Internal Revenue Code Section 382 ("IRC Section 382"). Events which may cause limitations in the amount of the net operating losses that the Company may use in any one year include, but are not limited to, a cumulative ownership change of more than 50% over a three-year period. Utilization of the federal and state net operating losses may be subject to substantial annual limitation due to the ownership change limitations provided by the IRC Section 382 and similar state provisions.

The Company's management believes that, based on a number of factors, it is more likely than not, that all or some portion of the deferred tax assets will not be realized; and accordingly, for the year ended December 31, 2019, the Company has provided a valuation allowance against the Company's U.S. net deferred tax assets. The net change in the valuation allowance for the years ended December 31, 2019 and 2018 was an increase of $86.0 million, $18.2 million, respectively.